Notes on Brams, "Prisoner's dilemma and the theory of metagames"

The prisoner's dilemma

There are a lot of different kinds of two-person (or two-party) interactions that count as prisoners' dilemmas. (In the prisoners' dilemma that is about prisoners, the two players who have been arrested *confess* or *do not confess*. In the general prisoners' dilemma, the players *cooperate* or *defect*.)

A prisoners' dilemma will always have a Nash equilibrium that is worse for both players than the outcome both could achieve if they cooperated, but the structure of the game prevents them from cooperating. As Brams explains at the bottom of p. 31, we can call the payoff that both players get if they cooperate R (for *reward*) and the payoff that both get if they don't cooperate (that is, if they defect) is P (for *punishment*). The two remaining payoffs are T (for *temptation* to double-cross the other player) and S (for *sucker's payoff*).

A player gets T when he or she defects and the other player cooperates.

A player gets S when he or she cooperates and the other player defects.

A game is a prisoners' dilemma when $T > R > P > S$. Brams put these variables in a table in figure 1.11 on p. 32.

Here are some additional guidelines for identifying a prisoner's dilemma.

A prisoners' dilemma can be set up either of these two ways.

	cooperate	**defect**
cooperate		
defect		

	defect	**cooperate**
defect		
cooperate		

The payoffs for the (cooperate, cooperate) outcome and the (defect, defect) outcome will be the same for both players. Keeping in mind that $R > P$, a prisoners' dilemma needs to match one or the other of these tables:

	C1	C2
R1	(R, R)	
R2		(P, P)

	C1	C2
R1	(P, P)	
R2		(R, R)

Then, the two remaining outcomes both have to contain the highest and lowest payoffs: (T, S) and (S, T). And they have to be in the right spots. To figure out where that is, find (R, R).

(1) From (R, R), if player 2 doesn't change, player 1 can switch strategies and make the outcome (T, S).

	C1	C2
R1	(R, R)	
R2	(T, S)	(P, P)

	C1	C2
R1	(P, P)	(T, S)
R2		(R, R)

(2) Similarly, from (R, R), if player 1 doesn't change, player 2 can switch strategies and make the outcome (S, T).

	C1	C2
R1	(R, R)	(S, T)
R2		(P, P)

	C1	C2
R1	(P, P)	
R2	(S, T)	(R, R)

The theory of metagames

A metagame is one that might be played just once, but there's the possibility that it will be played repeatedly (but with no particular known end point). And just like in a regular game, in a metagame, both players act at the same time (or around the same time) without knowing how the other player has acted. But in a metagame, after both players have acted, a moment is needed to figure out what's actually happened. This brings us to how the players act: instead of

selecting a strategy, both choose a metastrategy. A *metastrategy* is "a strategy for selecting a strategy" (bot., p. 34).

In the way that Brams sets up the metagame, player 2 has four metastrategies. (C stands for *cooperate*, D stands for *defect*.)

1. Choose C, regardless of what player 1 chooses.
2. Choose D, regardless of what player 1 chooses.
3. Tit-for-tat: choose C if player 1 chooses C, and choose D if player 1 chooses D.
4. Tat-for-tit: choose C if player 1 chooses D, and choose D if player 1 chooses C.

Player 1 can't have the same metastrategies because if both played tit-for-tat or tat-for-tit, nothing would get chosen. So, player 1's metastrategies are every combination of *cooperate* and *defect* in response to player 2's strategies. For example, these are player 1's first four metastrategies:

C/C/C/C: play C if player 2 plays any of his metastrategies.

C/C/C/D: play C if player 2 plays 1, 2, or 3, and D if he plays 4 (tat-for-tit).

C/C/D/C: play C if player 2 plays 1, 2, or 4, and D if he plays 3 (tit-for-tat).

C/D/C/C: play C if player 2 plays 1, 3, or 4, and D if he plays 2 (D regardless).

and so on for 12 more metastrategies.

Once the metastrategies are defined, we create the table using the payoffs given in figure 1.10. This metagame is just an expansion of that game, and so each payoff in the metagame is determined by those four pairs of strategies. For example, in the first column in figure 1.13, player 2 is playing C (no matter what player 1 does). And in response to *C always*, player 1 is playing whatever is in the first spot of _ /_ /_ /_ /. In the first four rows, that's C. So, both players would play *cooperate* and the payoffs are (-1, -1). On the fifth row, player 1 plays D in response to *C always*, and so, (D, C) yields (0, -10).

	C *Regardless*
1. C/C/C/C	(−1, −1)
2. C/C/C/D	(−1, −1)
3. C/C/D/C	(−1, −1)
4. C/D/C/C	(−1, −1)
5. D/C/C/C	(0, −10)

To take another example, player 2's tit-for-tat is an instruction to match whatever player 1 does. In response to *tit-for-tat*, player 1 is playing whatever is in the third spot of _ /_ /_ /_ /. So, if player 1 plays C, then so does player 2 and the payoffs are (-1, -1). If player 1 plays D, then so does player 2 and the payoffs are (-5, -5).

	C *Regardless*	D *Regardless*	*Tit-for-tat*	*Tat-for-tit*
1. C/C/C/C	(−1, −1)	(−10, 0)	(−1, −1)	(−10, 0)
2. C/C/C/D	(−1, −1)	(−10, 0)	(−1, −1)	(0, −10)
3. C/C/D/C	(−1, −1)	(−10, 0)	(−5, −5)	(−10, 0)
4. C/D/C/C	(−1, −1)	(−5, −5)	(−1, −1)	(−10, 0)
5. D/C/C/C	(0, −10)	(−10, 0)	(−1, −1)	(−10, 0)

Playing a metagame

To play a metagame, imagine each player selecting a metastrategy, writing it down, and putting it in an envelope. Once both players have done this, an intermediary (or a computer program) looks at both metastrategies and determines the outcome.

The Nash equilibrium

So, what should the players do? The Nash equilibriums are found the same way that they are in a regular game. Each player's best responses to the other player's metastrategies are circled in this table. The Nash equilibriums correspond to the pairs of metastrategies with both outcomes circled.

	C Regardless	D Regardless	Tit-for-tat	Tat-for-tit
1. C/C/C/C	(−1, −1)	(−10, 0)	(−1, −1)	(−10, 0)
2. C/C/C/D	(−1, −1)	(−10, 0)	(−1, −1)	(0, −10)
3. C/C/D/C	(−1, −1)	(−10, 0)	(−5, −5)	(−10, 0)
4. C/D/C/C	(−1, −1)	(−5, −5)	(−1, −1)	(−10, 0)
5. D/C/C/C	(0, −10)	(−10, 0)	(−1, −1)	(−10, 0)
6. C/C/D/D	(−1, −1)	(−10, 0)	(−5, −5)	(0, −10)
7. C/D/C/D	(−1, −1)	(−5, −5)	(−1, −1)	(0, −10)
8. D/C/C/D	(0, −10)	(−10, 0)	(−1, −1)	(0, −10)
9. C/D/D/C	(−1, −1)	(−5, −5)	(−5, −5)	(−10, 0)
10. D/C/D/C	(0, −10)	(−10, 0)	(−5, −5)	(−10, 0)
11. D/D/C/C	(0, −10)	(−5, −5)	(−1, −1)	(−10, 0)
12. C/D/D/D	(−1, −1)	(−5, −5)	(−5, −5)	(0, −10)
13. D/C/D/D	(0, −10)	(−10, 0)	(−5, −5)	(0, −10)
14. D/D/C/D	(0, −10)	(−5, −5)	(−1, −1)	(0, −10)
15. D/D/D/C	(0, −10)	(−5, −5)	(−5, −5)	(−10, 0)
16. D/D/D/D	(0, −10)	(−5, −5)	(−5, −5)	(0, −10)

You should read Brams's analysis on p. 37, but you can see in this table that, when playing this expanded version of the prisoners' dilemma with metastrategies, (cooperate, cooperate) is a Nash equilibrium.